A PRESHEAF WITH VALUES IN C IS A SHEAF EXACTLY WHEN ITS UNDERLYING PRESHEAF OF SETS IS

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Allowed standard prerequisites (editorial).

- The sheaf condition for presheaves with values in C is stated via the equalizer of a parallel pair, with F(U) mapping to the equalizer E (definition-sheaf-values-in-category). invoked at proof lines 1017-1020. Fixes the equalizer diagram and the map whose invertibility is equivalent to the sheaf condition.

Source reference. Original definition 0072: sources/stacks-sheaves/pinned-original-source.tex, source lines 936–936. Cited at proof lines 1017-1020 for the equalizer E and the diagram defining the sheaf condition in C; load-bearing inside the counted span.

Lemma 0.2. *Suppose the category $\mathcal{C}$ and the functor $F : \mathcal{C} \to Sets$ have the following properties:*

(1) *F is faithful,*
(2) *$\mathcal{C}$ has limits and F commutes with them, and*
(3) *the functor F reflects isomorphisms.*

Let X be a topological space. Let $\mathcal{F}$ be a presheaf with values in $\mathcal{C}$. Then $\mathcal{F}$ is a sheaf if and only if the underlying presheaf of sets is a sheaf.

Proof. Assume that $\mathcal{F}$ is a sheaf. Then $\mathcal{F}(U)$ is the equalizer of the diagram above and by assumption we see $F(\mathcal{F}(U))$ is the equalizer of the corresponding diagram of sets. Hence $F(\mathcal{F})$ is a sheaf of sets.

Assume that $F(\mathcal{F})$ is a sheaf. Let $E \in \mathrm{Ob}(\mathcal{C})$ be the equalizer of the two parallel arrows in Definition 0072. We get a canonical morphism $\mathcal{F}(U) \to E$, simply because $\mathcal{F}$ is a presheaf. By assumption, the induced map $F(\mathcal{F}(U)) \to F(E)$ is an isomorphism, because $F(E)$ is the equalizer of the corresponding diagram of sets. Hence we see $\mathcal{F}(U) \to E$ is an isomorphism by condition (3) of the lemma. □

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